

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Visualization Design Activity

Course Code:	DSA0502	Course Title:	Query Processing for Data Science
CO Assessed:	CO5	Assessment:	Tool 3– Visualization Design Activity
Weightage:	35%	Total Marks:	

CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication

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Section Batch: /	2014	Date of Submission:	20/08/2026
Faculty In-charge:	Dr. B.Shyamala Gowri	Project Mode:	Individual Submission

Code of Conduct

I Sandra Sudevan(192424422) (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: Sandrasudevan

No.	Activity	Task	Student Output	Time
1	Basic Plot Creation	Create a line plot using a small dataset of monthly sales. Add a title, x-axis label, y-axis label, legend, and grid.	Annotated line plot	15 min
2	Plot Component Challenge	Given an incomplete Matplotlib program, identify and add the missing title, labels, legend, markers, and grid.	Corrected Python program	10 min
3	Compare Visualization Types	Use the same dataset to create a line plot, bar plot, and scatter plot . Discuss which visualization communicates the data most effectively.	3 plots + justification	15 min
4	Pandas Visualization Activity	Create a DataFrame containing Product, Sales, and Profit. Generate appropriate visualizations directly using Pandas.	DataFrame + plots	15 min
5	Trend Visualization	Use monthly/weekly data to identify increasing, decreasing, and fluctuating trends using a line plot.	Trend visualization + analysis	15 min
6	Relationship Visualization	Plot two variables such as Advertising vs Sales using a scatter plot. Determine whether the relationship appears positive, negative, or weak.	Scatter plot + interpretation	15 min
7	Distribution Visualization	Create a histogram of student marks or employee salaries. Identify the approximate shape, center, spread, and possible skewness.	Histogram + observations	15 min
8	Frequency Visualization	Create a categorical dataset such as department-wise employee count. Represent the	Frequency chart	10 min

No.	Activity	Task	Student Output	Time
		frequencies using a bar chart.		
9	Scatter Matrix Visualization	Use a dataset containing at least four numerical variables. Generate a scatter matrix and identify the strongest visible relationships.	Scatter matrix + findings	20 min
10	Lag Plot Visualization	Use time-series data to create a lag plot. Examine whether the current observation appears related to the previous observation.	Lag plot + conclusion	15 min
11	Autocorrelation Visualization	Generate an autocorrelation plot for a time series. Identify lags with relatively strong positive or negative autocorrelation.	Autocorrelation plot + analysis	20 min
12	Bootstrap Visualization	Select a numerical dataset, repeatedly resample it, and examine the distribution of a chosen statistic such as the mean.	Bootstrap distribution + interpretation	25 min

3. Compare Visualization Types

Code:

```
# Dataset
month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun")
sales <- c(120, 150, 140, 180, 210, 190)
```

1. Line Plot

```
plot(month, sales,
      type = "o",
      xaxt = "n",
      main = "Monthly Sales - Line Plot",
      xlab = "Month",
      ylab = "Sales")
```

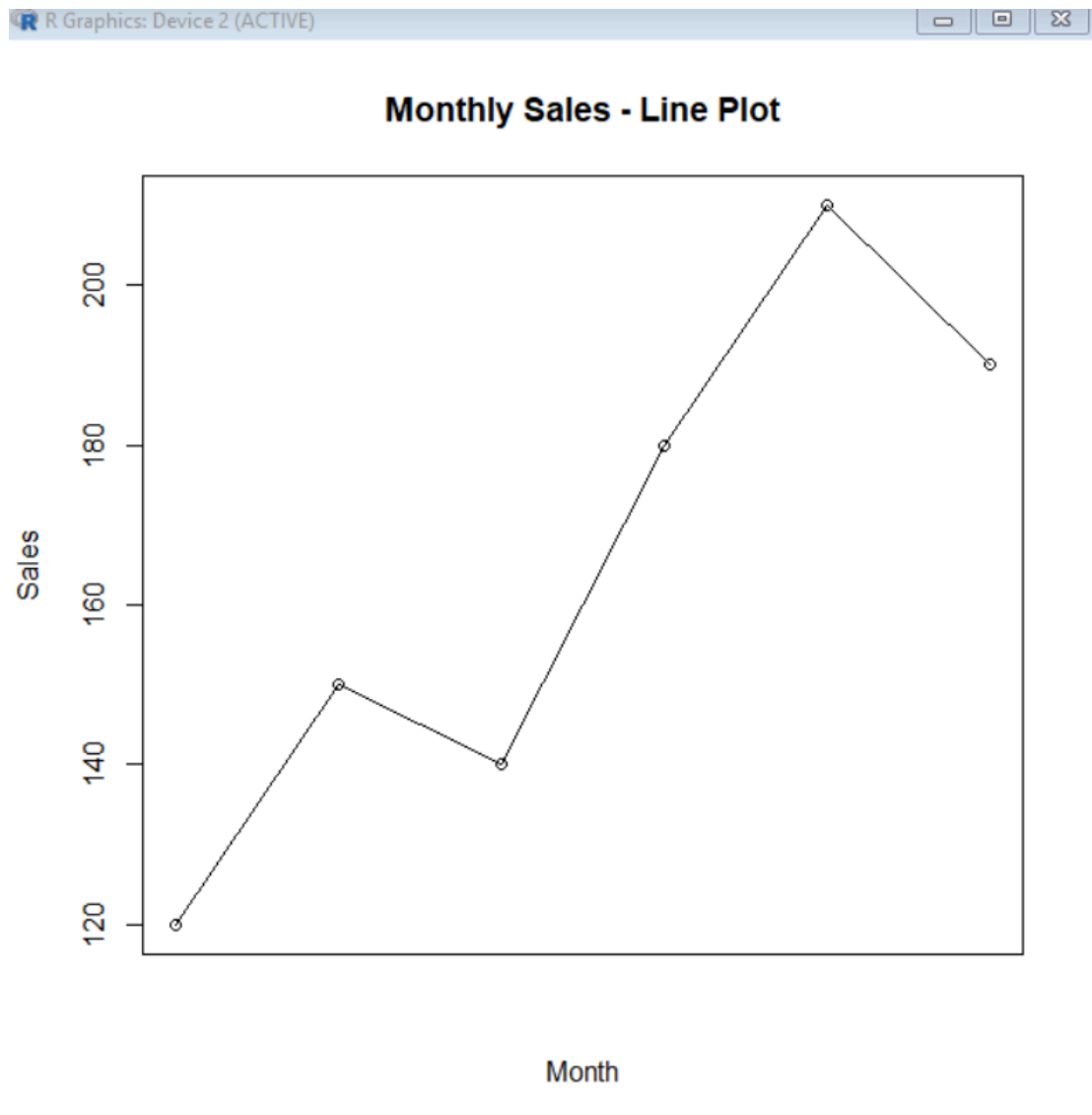
```
axis(1, at = month, labels = month_names)
```

2. Bar Plot

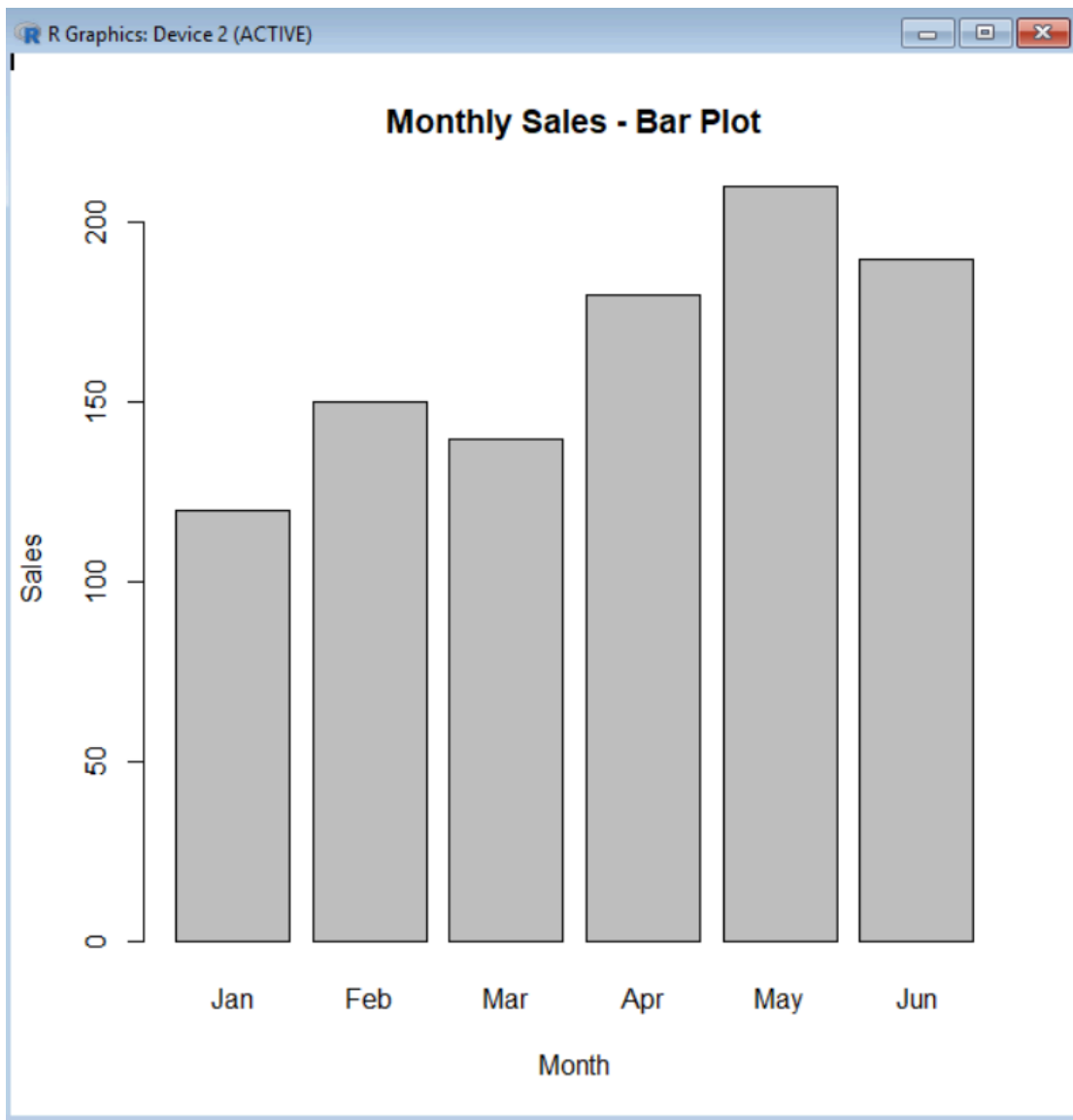
```
barplot(sales,  
        names.arg = month,  
        main = "Monthly Sales - Bar Plot",  
        xlab = "Month",  
        ylab = "Sales")
```

3. Scatter Plot

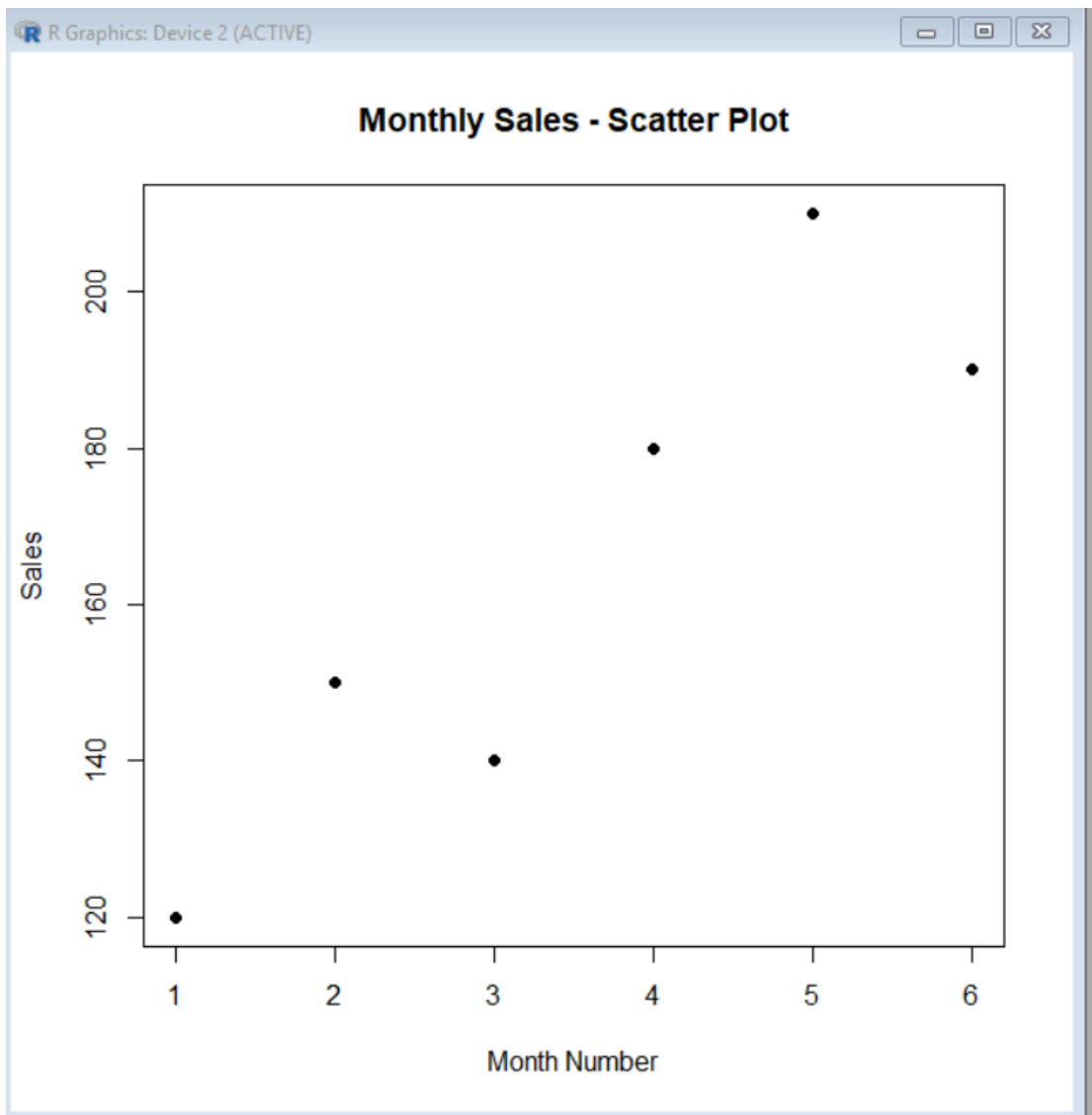
```
plot(sales,  
     main = "Monthly Sales - Scatter Plot",  
     xlab = "Month Number",  
     ylab = "Sales",  
     pch = 19)
```



Observation: Clearly shows the trend and changes over time.



Observation: Makes it easy to compare individual months and identify the highest sales.



A scatter plot shows the sales values as separate points. It is useful for seeing the values, but it is not very easy to understand the increase or decrease from one month to another.

1. Basic Plot Creation

Code:

```
# Monthly sales data
month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun")
sales <- c(120, 150, 140, 180, 210, 190)
```

Line plot

```
plot(sales,
     type = "o",
     xaxt = "n",
     main = "Monthly Sales",
     xlab = "Month",
     ylab = "Sales")
```

```
# Add month names  
axis(1, at = 1:6, labels = month)
```

```
# Add grid  
grid()
```

```
# Add legend  
legend("topleft",  
      legend = "Sales",  
      lty = 1,  
      pch = 1)
```

Title: Monthly Sales

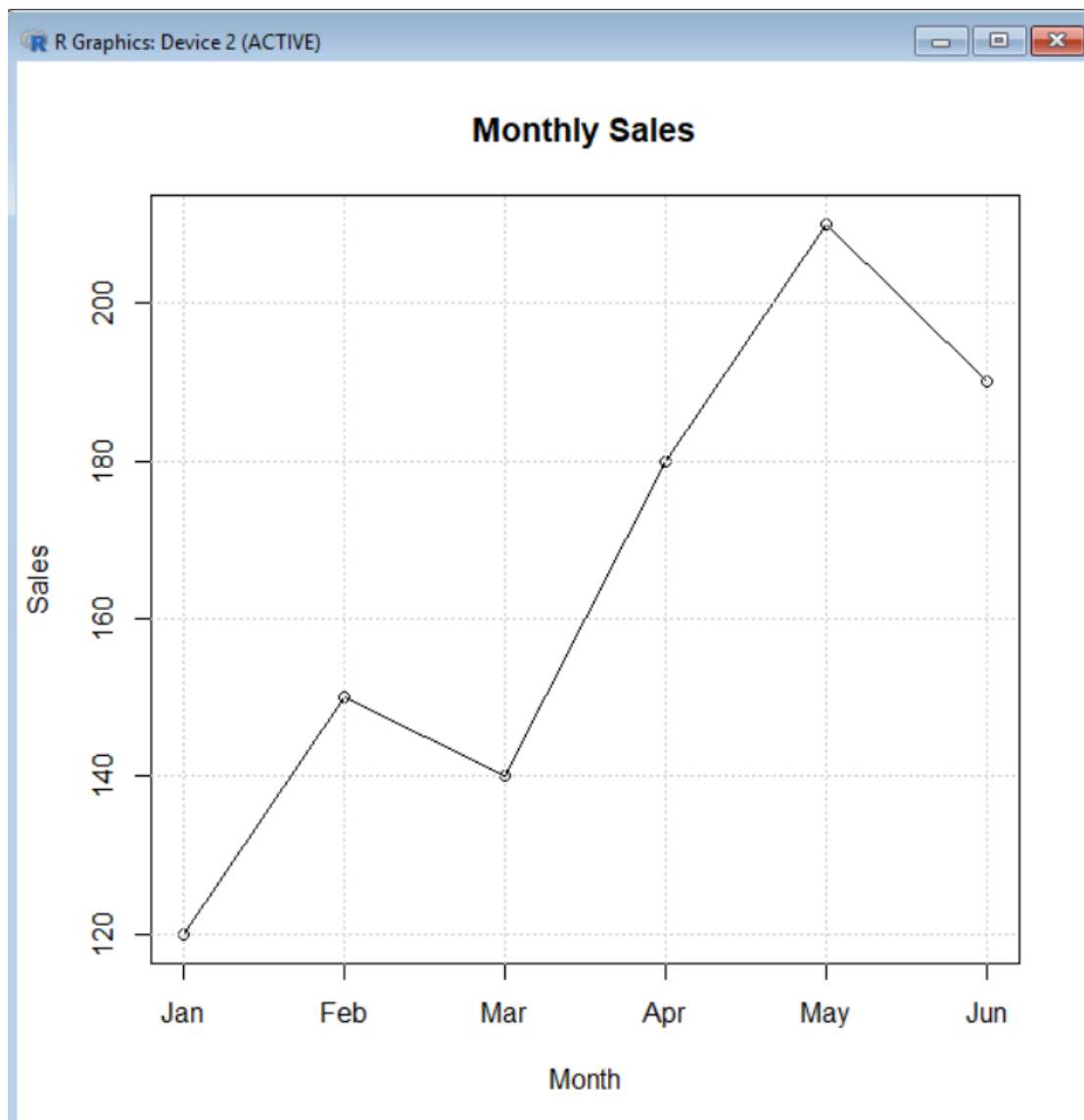
X-axis label: Month

Y-axis label: Sales

Legend: Sales

Grid: Makes the values easier to read

Line + points: Shows how sales change each month



The graph shows that sales increase from January to May. Sales reach the highest value of 210 in May and then decrease slightly to 190 in June.